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number 2

Part 1: Commutator and AC to DC conversion

1) P = 4

Mek. devir süresi = 2 second

θ_m = 2 degree

2.1) $f_m = \frac{1}{\text{Mekanik devir süresi}} = \frac{1}{2s} = 0,5\,Hz$ → Mechanical frequency

2.2) $f_e = f_m \times \left(\frac{P}{2}\right) = 0,5\,Hz \times \frac{4}{2} = 0,5 \times 2 = 1\,Hz$ → Electrical frequency

2.3) $T_e = \frac{1}{f_e} = \frac{1}{1\,Hz} = 1$ second → Electrical cycle

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3) The commutator acts as a mechanical rectifier or a rotary switch. It reverses the connection between the armature coil and the external circuit at the exact moment the current in the coil reverses its direction. This action ensures that the current flowing into the external circuit is always in the same direction, thus converting internal AC voltage to an external DC voltage.

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4) Equation formula

$\theta_e = \theta_m \times \left(\frac{P}{2}\right) \Rightarrow \theta_e = 2^\circ \times \left(\frac{4}{2}\right) = 2^\circ \times 2 = 4^\circ$

when shaft rotates by 2 mechanical degrees, coil travels 4 electrical degree.

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Part 2

$\Delta I = 2A$ (-1A to +1A

$T_C = 2$ second

$L = 2H$

1) $\frac{di}{dt} = \frac{\Delta I}{T_C} = \frac{2A}{2s} = 1 A/s$ => Average rate of change of current

2) $V_L = L \times \frac{di}{dt}$ = $V_L = 2H \times 1 A/s = 2V$ => Magnitude of the voltage (reactance volt.

3) The induced reactance voltage ($V_L = L \cdot di/dt$) as calculated, opposes the change in current according to Len's law. This opposing voltage prevents the current from reversing smoothly and completely within the brief commutation period. When the brush leaves the commutator segment, the current has not reached its final value, and this abrupt interruption of current flow causes a spark

4) 1) Interpoles (or Commutating Poles): Small poles placed between the main poles that generate a "commutating EMF" to directly oppose and cancel the reactance voltage.

2) Compensating Windings: Windings embedded in the main pole faces, used in large DC machines to counteract the effects of armature reaction, which also helps improve commutation.

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